AiSL

CEAS EXPO 2026

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https://aisl-production.up.railway.app/

Meet The Team!

Elshaddai Melaku

5th year CS and MBA student

Fareena Khan

5th year CS and a minor in Psychology student

Ikran Warsame

5th year CS and a minor in Women's, Gender, and Sexuality Studies student

Dr. Jillain Aurisano

Professor at the CS department at UC & our advisor

The Problem

- Learning ASL today is often **limited** to isolated vocabulary words
  - **Lack of resources** to learn ASL GLOSS, sentence structure
  - English text to sign tools use **SEE (Signed Exact English)**, not natural ASL
  - Learners memorize signs without learning fluency

Example

- **Literal (word-for-word English order):**
  - "I'm / going / to / the / store / tomorrow"
  - Literally this is awkward in ASL
- **More accurate ASL structure (time-topic-comment):**
  - "TOMORROW, STORE, I GO."
  - Why: ASL often sets **time first**, then the main idea.

The Gap

Why Current ASL Tools Fall Short

Structural Limitations

- Word-for-word English translation
- Produces Signed Exact English (SEE)
- Lacks ASL grammar teaching

Learning Depth

- Limited sentence-level learning
- Few solutions provide 3D playback
- Missing facial expressions in avatars

Integration Gaps

Almost none combine LLMs with grammar, custom avatars, and education.

Accessibility & Cost

Most advanced features are paywalled, preventing widespread adoption.

AiSL is not just Google Translate—it's building the infrastructure for true ASL comprehension.

Our Solution

AiSL

Smart Sentence Learning

AiSL helps users **learn full ASL sentences** in **proper grammar** through a **real-time signing avatar**.

Mission & Impact

Making ASL **accessible, inclusive, and culturally respectful**. By emphasizing authentic structure, we **bridge communication gaps** between Hearing and Deaf communities.

Software Architecture

Pipeline: Input text → NLP sentence restructuring → ASL GLOSS → sign database → Animation

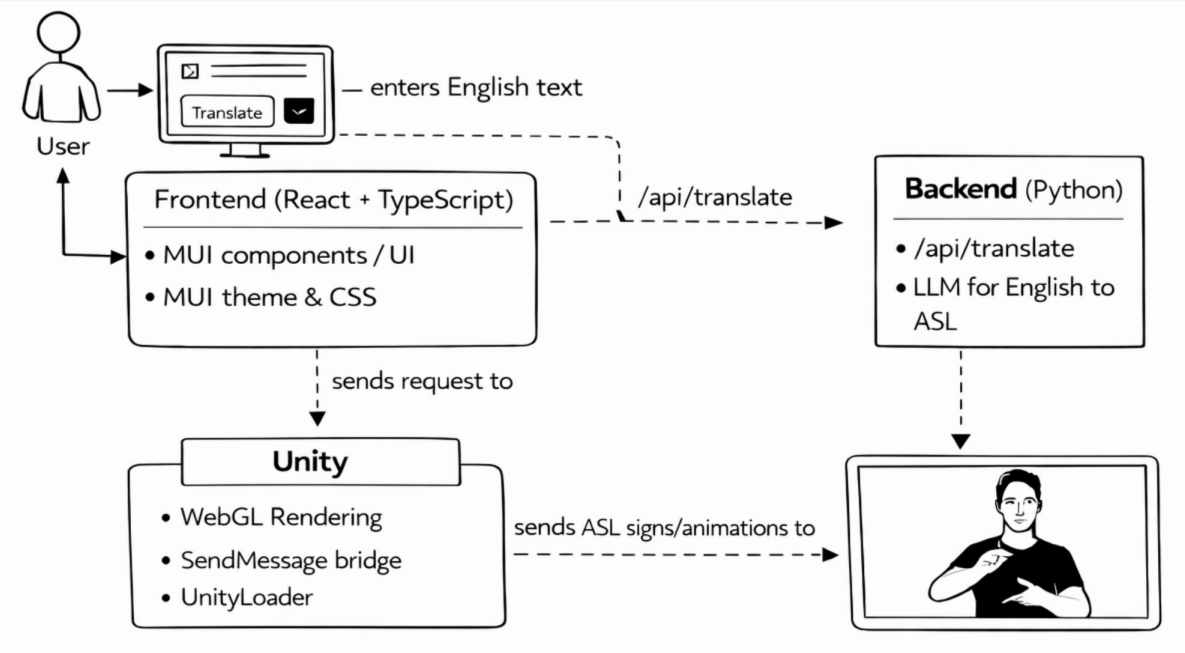
TS

React Vite

Material UI

WebGL

Unity



Flask

kaggle

python

OpenAI GPT-5.4-mini

3,915 signs!

Railway

3,915 signs

We support this many across our databases!

Engineering & Research Challenges

Limited Dataset

- Few ASL GLOSS/grammar datasets
- Limited public 3D sign libraries

Resources

- Limited sentence-level motion data
- Limited tools and research that can support sign motion

Under Researched

- 3D ASL generation is still emerging
- Few real-time signing systems exist

Funding

- No budget to professionally record our own signs
- Scaling requires
- GPU/cloud resources

Storage/Server

- ~~Train LLM~~ → Fine Tune
- Large animation assets
- High memory + rendering demands

Technical

- NLP + Unity + WebGL integration
- Smooth animation stitching was difficult

User Impact and User Interviews

UI Design

- ASL gloss / sentence output box
- ***Color-tagged grammar structure (subject, verb, object)**
- Dedicated learning resources page

Usability

- Adjustable playback speed and looping
- Easy to navigate

Translation

- Accurate individual sign output
- Strong sentence-level ASL translation
- ***Close-up avatar view improves learning clarity**

**User feedback of wanted implementation in bold*

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Future Additions

Short-term	Increase ASL sign dataset
	Enhance Unity Avatar to use facial expressions
	Migrate data to the Cloud for scalability
	Continue iteration on our VR prototype
Long-term	Train models to create AI-generated signs
	Add speech to text feature
	100% ASL fluency - Full ASL fluency: Non manual markers, classifiers, role-shifting - ***Arm Fluidity: smoother sentence flow** - BASL (Black American Sign Language) support: inclusive dialect - Facial grammar: higher realism and accuracy

Website DEMO

VR DEMO

Thank you! Any questions?

GitHub Repo

https://github.com/ikranw/AiSL

Live Website

https://aisl-production.up.railway.app/